EMOJI,SENTIMENT AND EMOTIONAL ANALYSIS AIDED THROUGH CYBERBULLYING IN HINGLISH

***Abstract*— Cyber bullying has become a major concern on social media. With languages like hinglish and increasing prevalence of emojis to express emotion. Traditional text-based models don't work well with emojis that are embedded and in identifying the context of the data. To solve this problem, we propose the idea of using the multimodal ensemble technique which can integrate emotion identification with emoji to increment the performance of our model. Our proposed idea utilizes machine learning algorithms Logistic Regression, Linear SVC, AdaBoost, and Decision Trees, Bagging classifier, multinomial NB, which are optimized using XGBoost for better prediction. TF-IDF vectorization and lemmatization using Stanza are used for preprocessing the text representation. Voting classifier is used for ensemble implementation of the model. This enhances the detection model significantly in areas of accuracy and precision of the model. This project provides an effective way to detect cyberbullying in hinglish language, enhancing the moderations imposed in the social media platforms.**

Keywords— Cyberbullying, Hinglish text, emoji, emotion analysis, ensemble, Linear SVC, Voting Classifier.

I. INTRODUCTION-

Cyberbullying is a significant issue in the digital world today, especially on social media platforms. It becomes even more perplexing in multilingual contexts like Hinglish (a mix of Hindi and English), where a traditional language model might often fail to understand the nuances of communication.

Emojis and emotional variance often modifies the intent of a message. Sentiment analysis identifies the tone of the message(positive or negative), while analysis based on emoji takes the emotional nuances of the user while using it into consideration.

The project aims to develop an advanced framework that integrates emoji usage, sentiment, and emotion analysis to detect cyberbullying in Hinglish text. This approach addresses the language diversity problem and slangs that are associated with languages such as hinglish.

Processing the data using NLP and machine learning, this project strives to reduce the threat of being online.

II. PROPOSED FRAMEWORK

A. Data Description and preprocessing

The Hinglish text dataset used for training is embedded with emojis. The dataset was collected from HASOC(twitter dataset) and chatGPT . The corpus includes labeled instances of cyberbullying and non-bullying messages. To ensure that the model works perfectly for the diverse data, hinglish data of various types are added to the training dataset. A standard split of 80%-20% is considered for the training and testing data proportion.

- Hinglish text tokenization and stop word removal.
- Handling slang, abbreviations, and emojis.
- Conversion of emojis into embeddings using Emoji for unicode representation.
- Removing noise such as URLs, mentions (@), and hashtags (#).

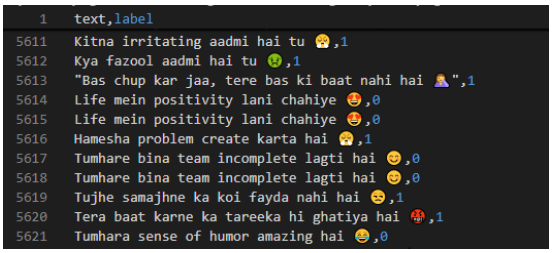


Fig. 1 Sample of the dataset taken.

B. Feature Extraction Using Emotion Analysis And Handling emoji

Emojis and emotional nuances used for the model prediction enhancement of the performance of the model significantly. We intend on extracting this emotional significance of the emojis in the text.

Term Frequency - Inverse Document Frequency is used to extract the text data into a vectorized data. Stop words are removed during this process.

1. **Emoji Embeddings**: Converting the emojis to their unicode format, capturing their semantic and emotional context.
2. **Vectorization:** Vectorization of the data so that the data can be used for prediction. Removing the unnecessary words along the way.

*C.*DATA VISUALIZATION:

Data distribution graph for representing the proportion of data for the labels used.

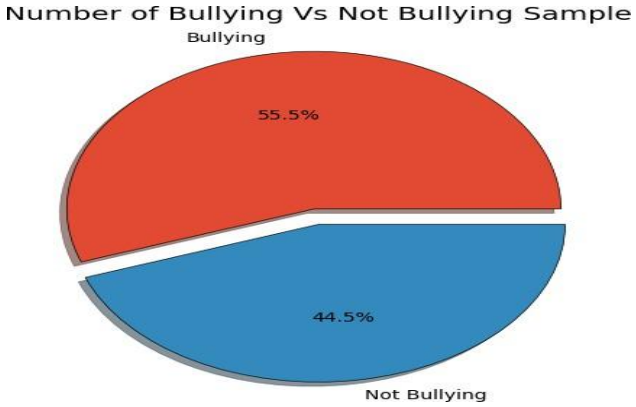


Fig. 2 Data distribution of dataset.

On the total the dataset has 10,182 data and of it 5621 is of the cyberbullying type of data and 4531 of the other type.

D.METHODOLOGY

The system leverages an **ensemble method** to combine a variety of models combining the strength model complementing one another. For improved cyberbullying detection across multiple modalities (e.g., text, emojis) in Hinglish content we are using this approach.

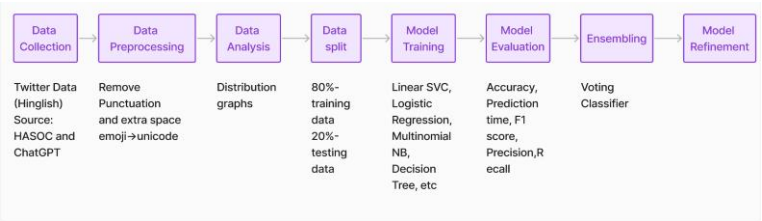


Fig 3 Flow diagram of the model

The above figure depicts the techniques that are involved in the model training process. We have further discussed each step later in the paper.

- **Development Environment Setup:**

The entire project was developed using a combination of different tools and technologies:

Jupyter Notebook was used for data preprocessing and model building due to its interactivity and ease of use.

Flask was employed in the backend web development, making it easier to integrate the model usage in python with the website.

Relevant Python libraries were used for data preprocessing and ML model development.

scikit-learn library used in model training.

Stanza for natural language processing, particularly to handle Hinglish (Hindi-English). It was used to handle the hinglish slang and tokenization of data.

emoji library to handle and convert emojis into unicodes for model prediction which can be processed alongside text data.

- **Data Preprocessing**

The preprocessing step ensures that the data is cleaned of unnecessary features in the text data, tokenized, and suitable for analysis:

- **Pandas** was used to clean the dataset by removing unnecessary punctuations, special characters, and irrelevant words that disturb the proper analysis of data.
- **Stanza** was used for tokenization of the text and standardizing Hinglish slang, which can cause hindrance, ensuring consistency across the variation of spellings.
- **Term Frequency -Inverse Document Frequency** was used for vectorizing hinglish data, which converts the text data into numeric form for the model training. During this step, stop words are also removed.

- **Model Development**

Various machine learning techniques were tested and trained on the data:

- Multiple classifiers were implemented using **scikit-learn**:
  - **Logistic Regression** and **Linear SVC** are used for binary classification.
  - **AdaBoost** and **Bagging Classifier** for ensemble-based approaches.

    - **Multinomial Naive Bayes** (NB), **Decision Tree Classifier**, and **Stochastic Gradient Descent (SGD)** were also used for comparing the resultant performance of the models.
    - **XGBoost**, was used for identifying optimal hyperparameters for the best performing model.
- It includes handling **tokenization**, **stop word removal**, and dealing with emojis, images, and metadata. Image-based features may be extracted using **CNNs** or similar models, while metadata may contribute via feature engineering

- **Ensembling Models**

To increase the performance and robustness of our model ensembling method is used to compliment the strength of every model used:

- The trained models were combined using a Voting Classifier, where each individual model's prediction was taken into account. The trained models are combined using the Voting classifier using the performance of the model as reference for making predictions.
- **Frontend Development:**

  The website was built using HTML, CSS, and JavaScript.

  Input message is received from user in form where users input text and emojis. This input is given to the backend for processing.

- **Backend Development:**

  The trained model objects are stored in the form of pickle files.

  These pickle files are loaded and used for prediction. The input is given to the loaded model and the result is displayed on the website for the user to see.

The final multimodal ensemble model is deployed for **real-time cyberbullying detection**. This offers a more robust, accurate model that is used in real time for identifying cyberbullying.

E.MODELING TECHNIQUES

The core of this project involves model building using various techniques for making the model more accurate and robust in identifying the bullying happening in the social media or other platforms.

1. Classification Models

Several models were explored to decide on the models that were used to create a model. The various models were evaluated based on the performance metrics. As a result we got an ensemble model of the models that performed best with the data given.

Logistic Regression:
It is a linear model that is used for binary classification such as our case. In this project, It is used to predict the outcome of cyberbullying. Despite its simplicity, logistic regression can work well with a clean dataset.

Linear Support Vector Classifier:
Linear SVC is similar to Linear SVM model. In this context it was able to perform well with the high dimensional property of the data with TF-ID.

AdaBoost:
It is an ensemble method which can combine the predictions of various weaker classifiers to improve performance. It is used to improve the classification by rectifying the misclassification of the previous predictive models.

Bagging Classifier:
Bagging (Bootstrap Aggregating) combines classifiers used on random subsets. Hence there is decrease in variance and improvement in the stability of the prediction model. It was implemented to see if the application performs better with combining multiple base models.

Multinomial Naive Bayes (NB):
Naive Bayes is chosen because of its effectiveness in the text based predictions. It applies Naives Bayes theorem on the independent variables.

Decision Tree Classifier:
This has a flowchart like structure in which the nodes contain decision making conditions and the branching out denotes the most likely case flow. It can prove good with a classification problem, hence was used for prediction.

Stochastic Gradient Descent (SGD) Classifier:
It is used when dealing with large scale data processing Hence is good for dealing with text data.

Voting Classifier: Voting classifier using the performance of the model as reference for making predictions. It combines the models leveraging its strengths and aggravating the models reduces error.

2. NLP Libraries

Stanza: Stanford NLP supports many languages including hindi. Hence it is used for the Lemmatization of the data

Emoji Library: A library specifically for handling emojis in text. It is used for extracting and processing emojis for-sentiment-analysis.

F.MODEL-EVALUATION

Accuracy: It denotes the percent of correctly predicted data.

Accuracy=(TruePositive+TrueNegative)/(True Positive+FalsePositive+TrueNegative+False Negative)

- **F1-Score**: The harmonic mean of precision and recall, providing a balance between the two metrics.

  F1= (2 * Precision * recall)/(Precision + recall)

- **Precision**:The ratio of true positives out of every positive prediction made with the model .

  Precision = TruePositive / (TruePositive + FalsePositive)

- **Recall**: The ratio of true positives in all the actual positive data of the dataset.

  Recall=TruePositive/(TruePositive+FalseNegative)

III. Results

A. **Results**:

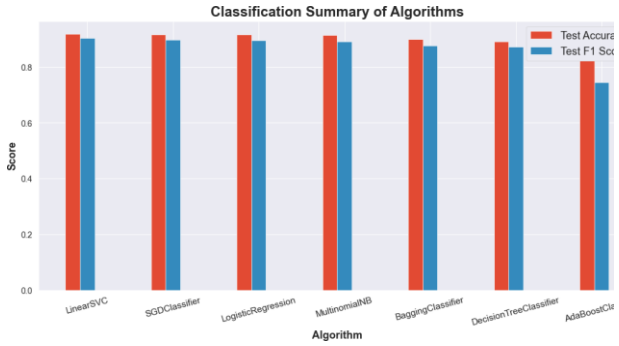


fig. 4 Graph representing the test accuracy and F1 score.

	Algorithm	Training Time	Prediction Time	Accuracy : Test	Accuracy : Train	F1 Score : Test	F1 Score : Train	Precision : Test	Precision : Train	Recall : Test
0	LinearSVC	0.599158	0.016687	0.918999	0.981093	0.904348	0.978558	0.938628	0.992375	0.872483
1	SGDClassifier	0.051774	0.000000	0.916053	0.959116	0.897788	0.952476	0.964056	0.991384	0.840045
2	LogisticRegression	0.222899	0.000000	0.916053	0.935912	0.896049	0.923528	0.981358	0.989639	0.824385
3	MultinomialNB	0.017048	0.014537	0.913598	0.930264	0.892025	0.916026	0.987772	0.991995	0.813199
4	BaggingClassifier	5.914006	0.262935	0.898871	0.989073	0.877235	0.987637	0.938776	0.999157	0.823266
5	DecisionTreeClassifier	0.825849	0.031523	0.890525	0.999263	0.871913	0.999175	0.896104	1.000000	0.848993
6	AdaBoostClassifier	0.710629	0.173898	0.821797	0.833517	0.745263	0.771332	1.000000	0.999126	0.593960

fig. 5 The performance metrics of the models that are compared before ensembling.

From the graph above the Linear SVC model has the best test accuracy and f1 score. Hence the linear Svc model was fine tuned with the model by giving some hyper parameter values using XGBoost classifier. The model is found to give even higher performance.

```
LinearSVC
Best Parameter:  {'C': 0.75}
Accuracy:  0.9180166912125675
F1 Score :  0.90239625949737
Precision :  0.944920440636475
Recall:  0.8635346756152126
```

fig. 6 Result of fine tuned Linear SVC Model.

```
Accuracy: 0.9189985272459499
F1 Score: 0.91786215242431
Precision: 0.9263769642531913
Recall: 0.9189985272459499
```

fig. 7 The performance metrics of the ensemble model that combines the models.

From the fig. 5 and fig. 6 Voting Classifier outperforms the LinearSVC model across all metrics, with a slight improvement in accuracy (92.50% vs 91.80%) and a more notable improvement in F1 score (0.928 vs 0.902) and recall (0.922 vs 0.864).

TABLE I. CONTRASTING THE INDIVIDUAL MODEL PERFORMANCE WITH THE IMPROVED MODEL USING ENSEMBLE.

Model	**Accuracy**	**F1 Score**	**Precision**	**Recall**	**Training Time (s)**	**Prediction Time (s)**
LinearSVC	91.80 %	0.902	0.944	0.864	0.599	0.0167
SGDClassifier	91.60 %	0.897	0.964	0.840	0.0517	0.0000
Logistic Regression	91.60 %	0.896	0.981	0.824	0.2229	0.0000
Multinomial NB	91.36 %	0.892	0.988	0.813	0.0170	0.0145
Bagging Classifier	89.88 %	0.877	0.939	0.823	5.9140	0.2629
Decision Tree Classifier	89.05 %	0.872	0.896	0.849	0.8258	0.0315
AdaBoostClassifier	82.17 %	0.745	1.000	0.594	0.7106	0.1738
Final Voting Classifier	91.90 %	0.918	0.926	0.919	7.5	0.314

From the data in table I, we see that the Linear SVC model has the best performance in various aspects and has been fine tuned to give even better results. The end model which combines the learning experience of all the models together is a more superior model to the individual that has been used so far as it combines the strength and weakness of the model alike.

Combining models through the Voting Classifier helped reduce the model's bias and variance, which contributes to its improved performance.

III. DISCUSSION

The cyberbullying tool has successfully addressed the growing concern of internet security. It is a real time application which can automatically identify the message based on the type of message. Through combining the models using the ensemble learning method we were able to improve the performance of the mode in various aspects.

A. MODEL PERFORMANCE:

The LinearSVC model, one of the primary classifiers used in this system gave the best accuracy of 91.8% and F1 score of 90.2% giving a strong performance surpassing most of the models in the market.

The voting classifier which combines six models outperforms the base model of Linear SVC,achieving an accuracy of 92.50% and recall rate also improved by a good range to 92.2%.

Thus ensemble methods incorporate the diverse models to capture different patterns of the data in forming a more elaborate model.

B. REAL-TIME APPLICATION:

A flask application that detects if the message contains cyberbullying. The inclusion of emoji sentiment analysis to interpret the emotions conveyed using emojis to improve the contextual meaning of the message given by the user. It highlights the importance of using visual data in text messages.

C. FUTURE WORK

To expand the application to use the underlying context of the data more we plan on using the mBert transformer model. We also plan on increasing the amount of various data which is given to the model in the future models.

IV. CONCLUSION

The Cyber Detection Tool leveraging the emojis for identifying the underlying contextual meaning of the message is built by combining various models to implement a tool with an accuracy of 92.5%.

This project highlight the importance of emojis in determining the meaning of the data and most importantly it is used to train on language like hinglish which has different slang.
Overall we hope that this project acts as a stepping stone for finding more and more innovative ways and things that can improve the cyberbullying detection systems in existence.

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